

REGN NO: FA25-BCS-017

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COURSE TITLE: DATA STRUCTURE LAB

LAB TASK:02

Q no 01:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int var = 10;
```

```
    // declare pointer and store address of var
```

```
    int* ptr = &var;
```

```
    // print value and address
```

```
    cout << "Value of var: " << var << endl;
```

```
    cout << "Address of var: " << &var << endl;
```

```
    cout << "Value stored in pointer ptr: " << ptr << endl;
```

```
    cout << "Value pointed to by ptr: " << *ptr << endl;
```

```
    return 0;
```

```
}
```

```
C:\Users\HP\OneDrive\Desktop >
Value of var: 10
Address of var: 0x6ffe34
Value stored in pointer ptr: 0x6ffe34
Value pointed to by ptr: 10

-----
Process exited after 0.3237 seconds with return value 0
Press any key to continue . . .
```

Q NO 2:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int var = 10;
```

```
    // Store the address of
```

```
    // var variable
```

```
    int* ptr = &var;
```

```
    // Access value using (*)
```

```
    // operator
```

```
    cout << *ptr;
```

```
    return 0;
```

```
}
```

```
C:\Users\HP\OneDrive\Desktop X + v
10
-----
Process exited after 0.3186 seconds with return value 0
Press any key to continue . . . |
```

Q no 3:

```
#include <iostream>

using namespace std;

int main() {

    int var = 10;

    // Store the address of
    // var variable
    int* ptr = &var;

    // Access the address value
    cout << ptr;

    return 0;

}
```

```
C:\Users\HP\OneDrive\Desktop X + v
0x6ffe44
-----
Process exited after 0.383 seconds with return value 0
Press any key to continue . . . |
```

Q NO 04:

```
#include <iostream>

using namespace std;
```

```

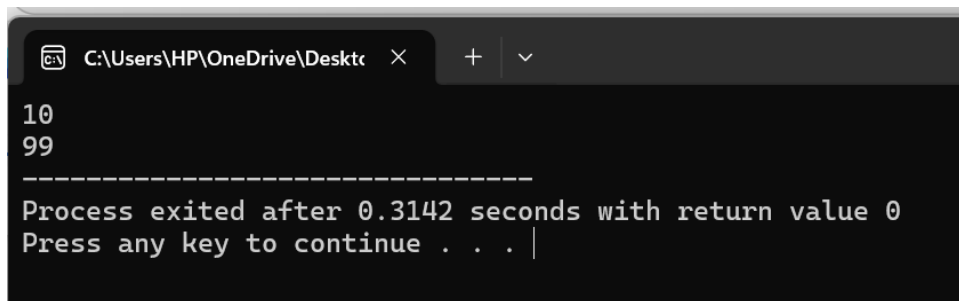
int main() {
    int a = 10;
    int b = 99;

    int *ptr = &a;
    cout << *ptr << endl;

    // Changing the address stored
    ptr = &b;
    cout << *ptr;

    return 0;
}

```



```

C:\Users\HP\OneDrive\Desktop
10
99
-----
Process exited after 0.3142 seconds with return value 0
Press any key to continue . . .

```

Q NO 05:

```

#include <iostream>

using namespace std;

int main() {
    int *ptr1;
    char *ptr2;

```

```

// Finding size using sizeof()

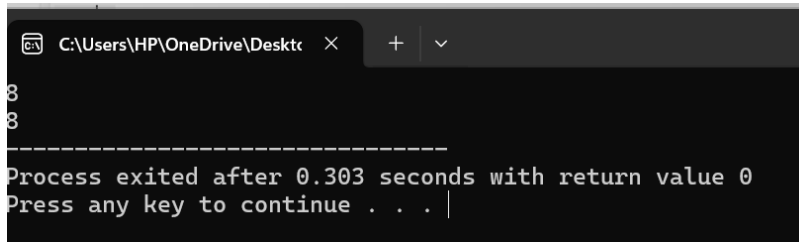
cout << sizeof(ptr1) << endl;

cout << sizeof(ptr2);


return 0;

}

```



```

C:\Users\HP\OneDrive\Desktop>
8
8
-----
Process exited after 0.303 seconds with return value 0
Press any key to continue . . .

```

Q NO 06:

```

#include <iostream>

using namespace std;

```

```

int main() {

```

```

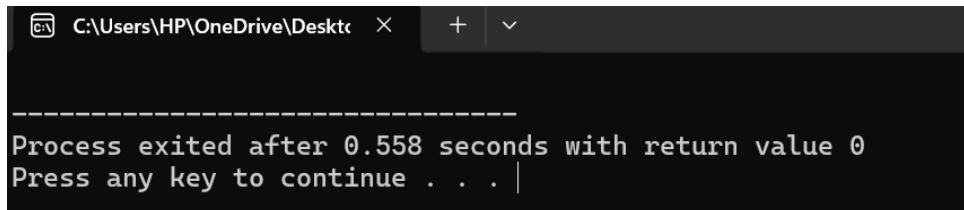
// Wild pointer

int *ptr;

return 0;

}

```



```

C:\Users\HP\OneDrive\Desktop>
-----
Process exited after 0.558 seconds with return value 0
Press any key to continue . . .

```

Q NO 07:

```
#include <iostream>

using namespace std;
```

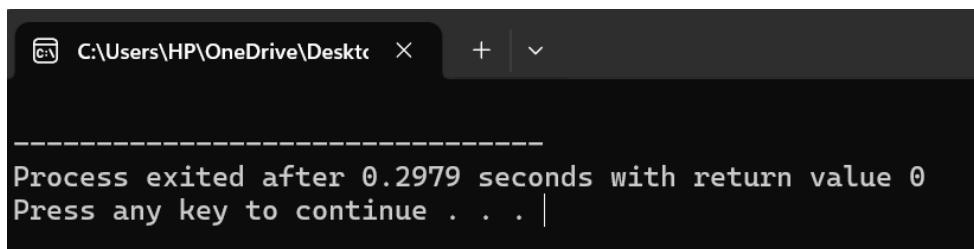
```
int main() {

    // nullptr pointer

    int *ptr = nullptr;

    return 0;

}
```



```
-----
Process exited after 0.2979 seconds with return value 0
Press any key to continue . . . |
```

Q NO 08:

```
#include <iostream> using namespace std;

int main() { int x = 42; // void pointer holding address of an int
void* ptr = &x;

// Error: cannot dereference void pointer
// cout << *ptr;

// Typecast before dereferencing
cout << "Value pointed by void pointer: " << *(static_cast<int*>(ptr)) << endl;

return 0;

}
```

```
C:\Users\HP\OneDrive\Desktop >
Value pointed by void pointer: 42
-----
Process exited after 0.6626 seconds with return value 0
Press any key to continue . . . |
```

Q NO 09:

```
#include <iostream>

using namespace std;

int main()
{ int var = 10;

// Store the address of
// var variable int* ptr1 = &var;int** ptr2 = &ptr1;

// Access values using (*)// operator

cout << *ptr1 << endl;cout << **ptr2;return 0;}
```

```
C:\Users\HP\OneDrive\Desktop >
10
10
-----
Process exited after 0.3952 seconds with return value 0
Press any key to continue . . . |
```

REFERENCES

```
#include <iostream>

using namespace std;

int main() {

    int x = 10;
```

```

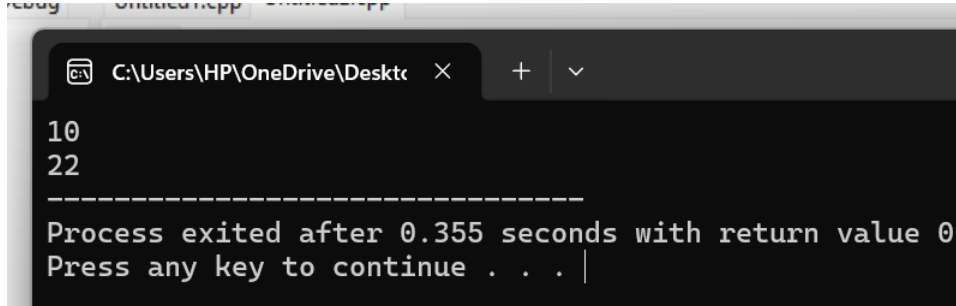
// ref is a reference to x.
int& ref = x;

// printing value using ref
cout << ref << endl;

// Changing the value and printing again
ref = 22;
cout << ref;

return 0;
}

```



```

C:\Users\HP\OneDrive\Desktop >
10
22
-----
Process exited after 0.355 seconds with return value 0
Press any key to continue . . .

```

Q NO 02;

```

#include <iostream>

using namespace std;

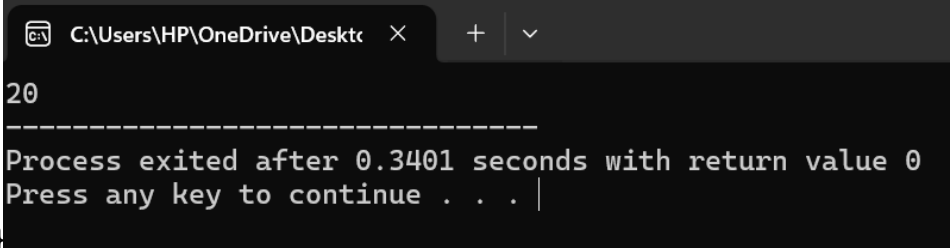
void modifyValue(int &x) {

    // Modifies the original variable
    x = 20;
}

```

```
}
```

```
int main() {  
    int a = 10;  
    // Pass a by reference  
    modifyValue(a);  
    cout << a;  
    return 0;  
}
```



```
C:\Users\HP\OneDrive\Desktop  X  +  v  
20  
-----  
Process exited after 0.3401 seconds with return value 0  
Press any key to continue . . . |
```

Q NO 03:

```
#include <iostream>  
using namespace std;
```

```
int& getMax(int &a, int &b) {  
  
    // Return the larger of the two numbers  
    return (a > b) ? a : b;  
}
```

```
int main() {  
    int x = 10, y = 20;  
    int &maxVal = getMax(x, y);  
}
```

```

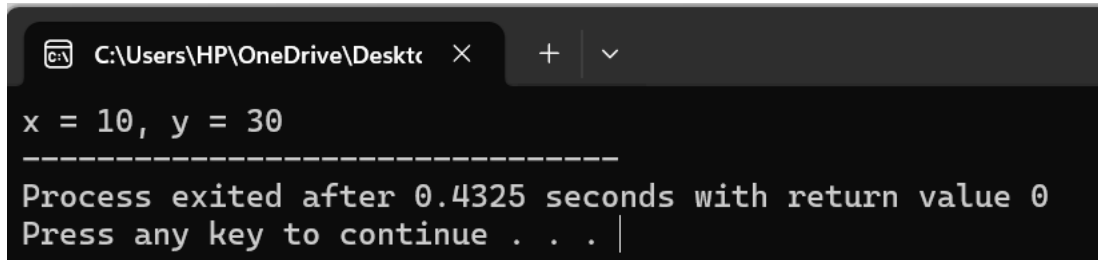
// Modify the value of the larger number

maxVal = 30;

cout << "x = " << x << ", y = " << y;

return 0;
}

```



```

C:\Users\HP\OneDrive\Desktop
x = 10, y = 30
-----
Process exited after 0.4325 seconds with return value 0
Press any key to continue . . . |

```

Q NO 04:

```

#include <iostream>

#include <vector>

using namespace std;

int main() {

    vector<int> vect{ 10, 20, 30, 40 };

    // We can modify elements if we
    // use reference
    for (int& x : vect) {

        x = x + 5;

    }

    // Printing elements
    for (int x : vect) {

        cout << x << " ";
    }
}

```

```
}
```

```
return 0;
```

```
C:\Users\HP\OneDrive\Desktop > .\Program.exe  
15 25 35 45  
-----  
Process exited after 0.3848 seconds with return value 0  
Press any key to continue . . .
```